

VALDEZ AP, AK, USA

WMO: 702756

Lat: 61.132N

Lon: 146.244W

Elev: 60

StdP: 14.66

Time Zone: -9.00 (NAK)

Period: 96-19

WBAN: 26479

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 1.4	(c) 6.5	(d) -17.9	(e) 2.1	(f) 12.9	(g) -14.6	(h) 2.5	(i) 15.7	(j) 32.6	(k) 23.4	(l) 26.6	(m) 26.8	(n) 5.2	(o) 90	(p) 0.680

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	70.3	56.4	66.2	55.3	63.7	54.1	58.5	66.9	56.8	64.6	55.3	62.4	8.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.4	63.1	61.4	53.5	61.0	60.0	52.0	57.7	58.3	25.3	67.0	24.2	64.5	23.4	62.2	64.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	17.3	12.4	-4.0	76.5	3.8	4.9	-6.7	80.0	-9.0	82.9	-11.1	85.7	-13.9	89.2
			-5.1	60.2	4.1	1.8	-8.0	61.5	-10.5	62.5	-12.8	63.6	-15.8	64.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.3	22.8	25.6	28.4	37.1	46.0	52.2	54.5	53.0	47.4	39.0	28.2	24.3		
	DBStd	13.10	9.35	7.65	6.54	4.77	5.10	3.87	3.39	3.49	3.78	5.04	7.75	8.14		
	HDD50	4639	844	684	671	388	145	16	2	6	92	340	655	796		
	HDD65	9759	1309	1104	1136	837	589	384	327	374	528	805	1105	1261		
	CDD50	357	0	0	0	1	21	83	140	98	14	0	0	0		
	CDD65	1	0	0	0	0	0	0	0	1	0	0	0	0		
	CDH74	36	0	0	0	0	0	14	8	14	0	0	0	0		
	CDH80	3	0	0	0	0	0	2	0	1	0	0	0	0		
	WSAvg	3.6	3.5	3.3	4.6	3.1	4.4	4.5	3.6	2.9	2.6	3.0	3.9	3.2		
(15) Precipitation	PrecAvg	65.9	5.9	5.6	4.4	3.5	3.0	2.8	3.7	6.4	8.7	8.3	5.7	7.2		
	PrecMax	93.3	15.2	14.2	10.0	8.1	6.7	6.1	9.0	18.2	16.7	15.4	20.6	17.3		
	PrecMin	34.8	0.0	0.6	0.8	0.6	0.7	0.1	0.2	0.4	0.7	0.3	0.4	1.3		
	PrecStd	12.8	3.8	3.4	2.6	2.0	1.4	1.4	1.9	3.9	3.9	4.0	4.3	4.3		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.0	41.4	45.4	57.1	68.4	75.1	73.8	75.1	69.1	54.4	45.7	39.5		
		MCWB	34.1	35.6	36.1	45.7	51.6	57.1	59.6	56.8	52.0	42.5	38.3	34.8		
	2%	DB	37.0	37.4	41.4	52.0	64.0	69.7	69.9	69.2	61.4	51.8	41.3	37.0		
		MCWB	34.0	33.3	34.8	41.7	49.6	56.3	58.6	55.7	49.7	42.0	36.6	34.7		
	5%	DB	36.0	36.3	38.8	47.7	60.8	64.3	65.6	64.3	57.1	48.1	38.7	36.0		
		MCWB	33.7	33.1	33.4	39.4	48.0	54.0	57.5	55.1	48.8	41.5	34.8	34.0		
	10%	DB	34.2	34.5	36.8	45.4	55.4	62.6	63.1	61.5	54.5	46.0	36.6	34.2		
		MCWB	32.7	32.3	32.5	38.6	46.2	53.1	55.6	53.7	48.4	40.7	33.4	32.7		
	0.4%	WB	37.5	37.0	39.4	46.2	53.6	59.5	61.4	60.1	53.7	47.4	41.0	37.4		
		MCDB	39.3	39.2	43.8	56.8	64.8	71.7	70.0	68.6	60.8	49.4	44.6	38.0		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.4	35.3	36.9	42.4	50.8	56.7	59.1	57.7	52.1	45.8	38.1	35.6		
		MCDB	36.2	36.7	39.5	48.9	61.4	66.7	67.1	65.6	58.1	47.4	40.2	36.3		
	5%	WB	34.1	34.1	35.2	40.7	49.1	54.9	57.4	56.1	50.9	44.1	36.3	34.2		
		MCDB	34.9	35.3	37.2	46.0	57.6	63.8	64.7	63.3	55.8	46.3	38.0	34.8		
	10%	WB	32.7	32.9	33.9	39.3	47.5	53.2	55.9	54.4	49.7	42.4	34.3	32.9		
		MCDB	33.7	34.2	36.0	44.6	55.7	61.1	62.6	60.1	53.9	45.3	35.9	33.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	10.4	10.8	11.3	13.0	13.9	12.9	13.0	11.7	9.6	8.4	8.9		
		MCDBR	9.7	10.7	12.5	17.1	22.1	22.3	20.7	20.6	18.5	13.9	9.5	9.6		
		MCWBR	7.6	8.1	8.8	10.4	11.7	12.9	13.2	12.9	11.2	8.6	6.5	7.4		
		MCDBR	7.6	7.8	9.4	14.0	18.1	19.7	17.9	17.5	13.8	9.4	7.3	7.3		
	5% WB	MCWBR	6.8	6.8	7.1	9.1	10.5	12.2	12.3	12.2	9.5	7.3	5.8	6.4		
(40) Clear-Sky Solar Irradiance	taub		0.257	0.315	0.322	0.386	0.400	0.406	0.402	0.401	0.343	0.318	0.270	0.247		
	taud		2.010	2.035	2.075	1.940	2.051	2.163	2.238	2.244	2.373	2.196	1.988	1.897		
	Ebn at Noon		139	198	246	249	258	259	257	246	243	199	125	80		
	Edh at Noon		13	25	34	48	46	43	38	35	26	21	13	9		
(44) All-Sky Solar Radiation	RadAvg		90	315	819	1396	1733	1785	1598	1217	784	359	129	49		
	RadStd		10	43	77	106	118	104	98	116	101	41	20	5		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page